



PHOTOMASK TUNING SOLUTIONS

ZEISS ForTune Systems

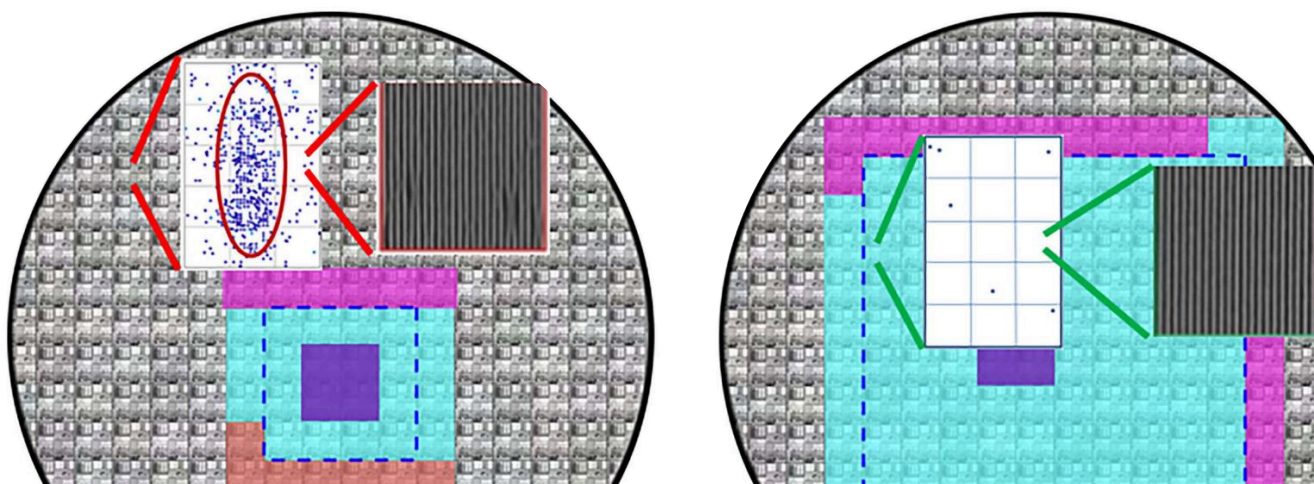
Exceeding the boundaries of lithography

Wafer Fabs highly depend on predictability and credibility. Achieving high yields is essential to convince their customers of the capability of the fab. Any process excursion that reduces the yield significantly undermines the trust of the customer into the fab's capability and reliability. ZEISS offers an innovative solution preventing excursion by improving intra-field Critical-Dimension-Uniformity (CDU) and improving the Wafer On-Product-Overlay (OPO).

Reduces process defects (DUV)

Improves Wafer On-Product Overlay (DUV)

Reduces Mix and Match Overlay (EUV)

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High performance mask tuning system

ZEISS ForTune and ZEISS ForTune EUV improve wafer intra-field litho parameters by mask tuning in high lateral resolution on top of any other available solution in the market and at low cost. The systems are available for all market segments (Memory, Logic in DUV and EUV) and cover the following applications:

CDC Application

The probability of process defects caused by wafer intra-field CDU can be reduced applying ZEISS ForTune. Following the excursion prevention strategy increases thereby wafer yield as can be seen in the exemplified images.

RegC

®

Application

Both, ForTune and ForTune EUV, are complimentary to the scanner. The RegC

®

functionality converts wafer intra-field OPO contributions from non correctable to correctable errors by the scanner in high lateral resolution. By that the wafer OPO is improved on top the scanner knobs capabilities.

ForTune

Intra-field high lateral resolution solutions for CDU & OVL

With its latest optical design ZEISS ForTune takes the two main tuning applications to the next level in terms of efficiency, accuracy and throughput.

An ultra-short pulse laser technology allows to change the optical and mechanical properties of the shading elements and thereby to improve wafer intra-field CDU and overlay.

ZEISS ForTune comprehensive software suite allows the users to predict the specific process performance reliably and accurately before the actual process execution.

Graphical outputs allow comparison of the initial problem with the predicted post process solution, including suitable statistics for the scanner compensation to be used later in HVM.

ForTune EUV

Improving wafer overlay for EUV technology

ZEISS ForTune EUV is a high-performance mask tuning system improving wafer overlay for EUV technology, based on mask registration or wafer intra-field overlay input.

The newly developed system enables mask tuning for EUV masks with low backside coating mask transmission (e.g. standard Hoya TaB 70nm backside coating) by well controlled deformation of the EUV mask bulk material.

Digital Solutions

For ForTune and ForTune EUV

A high volumen computational platform enables productivity and reliability enhancement for fast computation.

FTM - ForTune Tuning Module - The Tuning Module software package has been developed for ZEISS ForTune to facilitate recipe creation as well as extensive data analysis and simulations prior to the ForTune process (CDC & RegC® applications). This module's versatile job handling capabilities ensure high process efficiency, superior prediction accuracy and on-the-spot decision making.

[See all digital solutions](#)



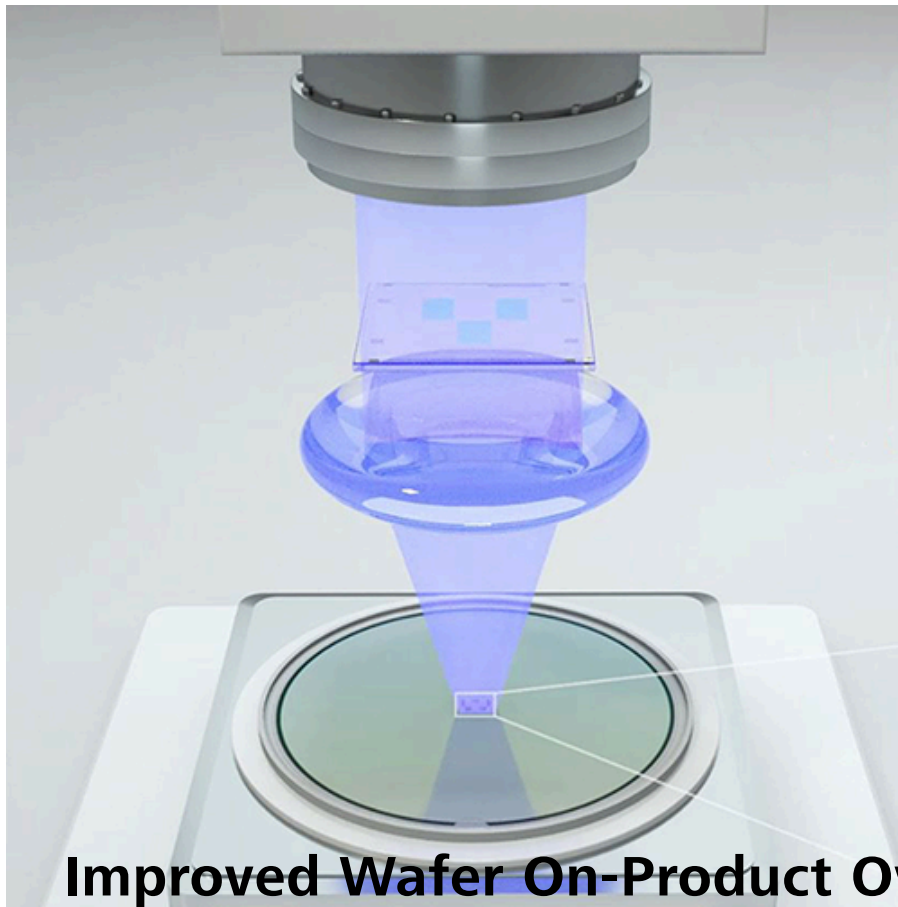
Reducing process defects ↕

By Excursion Prevention applying ForTune

Patterning excursion is referring to (multiple) parameters deviation from normal that leads to process out of the allowed tolerances. The patterning excursion can be caused by process parameters and lithography parameters like focus, mask CDU and dose. Patterning defects can be caused by interaction of multiple lithography parameters, even if the individual parameters are in spec.

To reduce wafer intra-field process defects Excursion Prevention by ZEISS ForTune system is a proven strategy. This can be achieved by using the CDC Application of ForTune: It enables excursion prevention at low effort and low cost in wafer fabs. As a result, yield loss will be reduced

significantly.



Improved Wafer On-Product Overlay ↕

Registration control functionality helps to achieve better OPO for DUV and EUV

The Wafer On-Product Overlay (OPO) is another limiting key parameter for chip performance which needs to be well controlled. The main contributed parameters can be categorized as inter-field and intra-field parameters. Both can be controlled by the scanner. However, the scanner is limited in the intra-field overlay resolution and can control only up to the 3rd order polynomial across the slit. This scanner limitation leads to Non-Correctable Errors (NCE) in high orders that cause OPO is out of specification. ForTune converts NCE to CE (Correctable Errors) resulting in an improved intra-field contribution. This helps to reduce the Wafer On-Product Overlay for DUV as well as for EUV masks.



Improved Mix and Match

ForTune EUV reduces Overlay between DUV and EUV mask layers and EUV to EUV layers

A high-end mask set consists of multiple DUV masks and some EUV masks. The various layers must be printed accurately on top of each other to achieve a functional and high performing microchip.

With its RegC

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functionality, ForTune EUV reduces the Intra-field contribution which leads to an optimized Wafer Mix & Match Overlay (MMO).

How does mask tuning work?

Watch the video about ZEISS ForTune technology

<https://www.zeiss.com/semiconductor-manufacturing-technology/products/photomask-solutions/mask-tuning.html>

Brochures



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